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A soft magnetoelastic ball for self-powered Parkinson's disease diagnosis

Trinny Tat,[1,2] Jing Xu,[1,2] Xiao Xiao,[1] Songyue Chen,[1] Zhaoqi Duan,[1] Xun Zhao,[1] Guorui Chen,[1] Junyi Yin,[1] and Jun Chen[1,3,*]

[1]Department of Bioengineering, University of California, Los Angeles, Los Angeles, CA 90095, USA

[2]These authors contributed equally

[3]Lead contact

*Correspondence: jun.chen@ucla.edu

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THE BIGGER PICTURE The soft magnetoelastic ball represents a quantitative diagnostic tool for Parkinson's disease (PD), complementing physicians' examinations of motor symptoms. PD manifests with motor symptoms such as slowness of movement (bradykinesia), resistance to movement (rigidity), and/or tremors. Current examination methods require a specialist to ask patients to rate their symptoms on a subjective scale from 0 (normal) to 5 (severe). However, by the time patients perceive noticeable differences, PD may have already advanced to a later stage. Subjectivity also introduces variability and limits early detection, highlighting the need for objective assessments.

The soft magnetoelastic ball addresses this need by providing a complementary, objective examination. The device leverages the magnetoelastic effect, which converts movements into magnetic fields, and magnetic induction, which transforms the magnetic fields into measurable electrical signals. Together, the waterproof and flexible design exploits these coupled effects to convert hand movements into electricity, enabling quantitative and objective detection of PD motor symptoms. Using machine learning algorithms, the device was able to differentiate between healthy and simulated PD hand movements with an average prediction accuracy of 98.36%. Additionally, a companion mobile application was developed to facilitate seamless data transfer to physicians or third parties, enhancing the diagnostic workflow. This work supports PD clinical evaluations by offering a quantitative tool that complements subjective procedures while improving accessibility for patients with limited technological proficiency or visual, verbal, hearing, or writing impairments.

SUMMARY

Parkinson's disease (PD) affects over 10 million people globally. Yet, its diagnosis, neuropathology, and treatment remain challenging. PD primarily impacts the elderly, often necessitating complex medication regimens and increasing the risk of medication errors and overdoses. In this study, we report a machine-learning-assisted soft magnetoelastic ball for PD diagnosis. Utilizing the magnetoelastic effect and magnetic induction, it converts hand tremors into high-fidelity electrical signals with a 50 ms response time and a 64.6 dB signal-to-noise ratio. With 300% stretchability and Young's modulus of 214.53 kPa, it maintains a pressure sensitivity as low as 0.95 kPa. The device underwent ten simulated PD features before implementation with a one-dimensional convolutional neural network to distinguish, achieving 98.36% classification accuracy. A customized smartphone application was integrated to record and transmit data to clinicians, enabling real-time, data-driven PD diagnostics. This inclusive tool enhances accessibility for individuals with limited technological proficiency or visual, verbal, hearing, or writing impairments.

INTRODUCTION

Parkinson's disease (PD) is a rapidly growing neurodegenerative disease, affecting nearly 10 million people and amounting to about $52 billion in total economic burden.[1] Unfortunately, PD remains in a terminal condition with no available treatments to reverse or halt the progression; the existing options merely aim to slow its course. Since PD targets mostly the elderly, who often take multiple medications, the risk of adverse drug reactions can be higher.[2,3] Correspondingly, the effectiveness of levodopa, a common medication for PD, can diminish over time as the disease progresses.[4–6] Therefore, providing appropriate quantification of disease progression and timely, personalized drug response in PD remains a major challenge for the PD therapeutic and diagnostic communities.

PD is often characterized by motor symptoms of bradykinesia and either tremors or rigidity.[7,8] Diagnosing these symptoms oftentimes requires a specialized physician to ask the patient to move certain body parts (e.g., hands, fingers, arms, head, and feet) as a part of their qualitative, feature-displaying evaluation. The most common PD rating scales—the Unified Parkinson Disease Rating Scale (UPDRS), with the Movement Disorder Society (MDS)-UPDRS as the newest revision[9]; Hoehn and Yahr stages[10]; and the Schwab and England rating of the activities of daily living scale[11]—rely heavily on patients' subjective inputs.[12,13] This subjectivity introduces variability and psychometric limitations, particularly in detecting early-stage PD symptoms and tracking subtle disease progression.[14] Advanced imaging techniques, including magnetic resonance imaging and DaTscan brain scans, can also confirm the diagnosis, even though these cumbersome scans are only extra steps following several consultations and qualitative evaluations and require a third-level medical referral.[15,16] Furthermore, since they can be invasive, side effects such as headache, nausea, vertigo, dry mouth, dizziness, hypersensitivity reaction, and pain may occur in PD patients.[17,18]

Given PD's impact on motor control and manual dexterity, many diagnostic software and hardware tools[19–21] have been explored to provide additional PD assessment. However, these devices often require patients to engage in unfamiliar or technology-dependent tasks on top of their standard clinical evaluation. In contrast, a handheld ball offers a uniquely intuitive and accessible form. The ball aligns with existing standard PD assessments, such as those in the Stanford Medicine 25 protocol, without introducing extra complex procedures and burden to a physician's workflow. Furthermore, it serves as an active rehabilitation equipment for at-home hand exercises,[22,23] helping manage manual dexterity impairment.

Traditionally associated with rigid metals and metal alloys since 1865, the magnetoelastic effect, defined as the change of a material's magnetic properties in response to mechanical stress or strain, was only recently discovered and demonstrated in soft materials in 2021.[24] These soft magnetoelastic materials, combining various material options for polymer matrices and magnetic particles,[25–27] have been investigated extensively for applications in assistive technologies,[28–32] sustainable energy,[33–38] and personalized healthcare,[21,39–45] owing to their large mechanical compliance and magnetomechanical coupling (MC). Leveraging this fundamentally new principle, a magnetoelastic generator (MEG) integrates the magnetoelastic effect from a magnetomechanical coupling layer that converts mechanical load into a magnetic field variation, with a magnetic induction (MI) layer that transduces these magnetic signal variations into readable electrical output. As each layer can assume various adaptive forms, the coupled system realizes a self-powered mechanism that achieves a mechanical-to-electrical conversion without the need for an external power supply. Beyond energy harvesting, early demonstrations of MEG-enabled self-powered sensors have predominantly focused on monitoring a diverse range of biomechanical activities.

In this study, we explore a soft magnetoelastic ball designed to quantitatively detect early PD motor symptoms. Possessing intrinsic waterproof, flexible, and ultrahigh-current output properties, the soft magnetoelastic ball can convert hand tremors, rigidity, and bradykinesia into high-fidelity electrical signals, offering an accurate and noninvasive diagnostic solution. The soft magnetoelastic ball holds a stretchability of 300%, a Young's modulus of 214.53 kPa, a pressure sensitivity as low as 0.95 kPa, a short response time of 0.05 s, and a signal-to-noise ratio (SNR) of 64.6 dB within the PD tremors frequency range of 4–6 Hz. To improve technical accessibility and streamline communications between patients and physicians, the device's raw electrical signals were analyzed using one-dimensional convoluted neural networks (1D-CNN) to determine healthy vs. simulated PD signals with an average prediction accuracy of 98.36%. To enhance its capabilities, we have developed a companion smartphone application that integrates seamlessly with the soft magnetoelastic ball, forming an intelligent PD diagnostic system. This platform not only complements physicians' PD motor evaluation with quantitative results recorded in real time but also supports at-home continuous monitoring of PD symptoms. Overall, the soft magnetoelastic ball represents a self-powered, cost-effective, and user-friendly technology poised to improve PD diagnostics and disease management.

RESULTS

Structural design of the soft magnetoelastic ball

The soft magnetoelastic ball was designed to provide objective measurements of PD motor manifestations, incorporating the traditional clinical evaluations that focus on hand and finger movements (Figures 1A and 1B). This device leverages the giant magnetoelastic effect in soft polymers and electromagnetic induction to convert subtle PD tremors into electrical signals.[24] Inspired by the elasticity of a common stress ball, the soft magnetoelastic ball consists of two main mechanisms: (1) an MC layer that converts mechanical into magnetic signals via the magnetoelastic effect and (2) a conductive MI layer that converts magnetic signal variations into electrical output via electromagnetic induction.

In detail, the soft magnetoelastic ball comprises four key components (Figure 1C). The outermost component is a commercially available soft silicone cover, acting as structural support while providing enough strain and stretch for collecting pressure-sensitive information. The two middle layers comprise a highly sensitive ball-shape MEG for mechanical-to-electrical conversion using the giant magnetoelastic effect and electromagnetic induction. The 6-mm-thick magnetoelastic wall, fabricated using spherical molds, acts as the MC layer. This MC layer was prepared by embedding commercially obtained neodymium-iron-boron (NdFeB) magnetic powders into a soft polymer matrix. Originally supplied as crushed ribbon powder, these uncoated particles exhibit platelet-like, faceted morphologies with rough surfaces, which likely enhance their adhesion to the matrix. The majority of the particles fall within the 45–89 μm size range. The 600-tightly-wound copper fibers, acting as the MI layer, fuse with the MC layer to facilitate the mechanical motion

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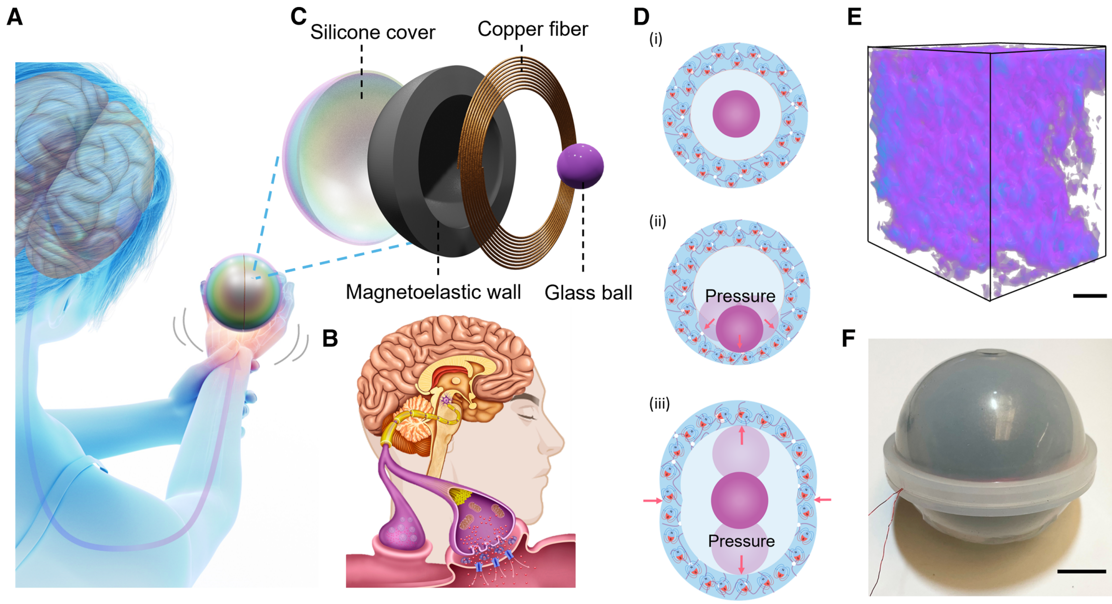

Figure 1. Illustration of a soft magnetoelastic ball and its giant magnetoelastic effect
(A) Schematic of a user holding the soft magnetoelastic ball.
(B) Overview of the underlying cause of PD.
(C) Deconstructed soft magnetoelastic ball that includes four components: a stretchable silicone cover, a magnetoelastic wall, a copper fiber coil, and a glass sphere.
(D) Sketches of the internal structure of the soft magnetoelastic wall while being acted upon by the glass sphere, showing a magnetic flux and micromagnets dispersing throughout the polymer silicone matrix.
(E) Micro-CT scans of the internal structure of the magnetoelastic wall. Scale bar: 50 μm.
(F) Photograph of the constructed soft magnetoelastic ball. Scale bar: 12 mm.

conversion into readable electrical signals. The final component, a glass sphere, plays a crucial role in inducing multidirectional stress on the MEG without the need for direct external force interaction.

When the glass sphere collides with the MEG, two-level interactions occur. At the microscale view, the deformed shape caused by the collision of the glass sphere induces magnetic particle-particle interactions (MPPI), owing to distance changes and rotation of the magnetic particles. At the atomic point of view, this collision also causes magnetic dipole-dipole interactions (MDDI), leading to rotation and movement variation of the magnetic dipole within the particles. Figure 1D shows the mechanism of that incident. To observe the initial state of magnetic particle alignment before deformation, the micro-computed tomography (micro-CT) technique was employed, as depicted in Figure 1E. By leveraging the principles of magnetoelasticity and electromagnetic induction, the soft magnetoelastic ball introduces an objective approach for detecting the subtle movements associated with PD. This technological innovation holds great potential for improving diagnosis and, in the future, for monitoring PD progression, providing healthcare professionals with objective and reliable measurement data for assessing the progression of the disease.

Magnetic and electrical properties

Prior to conducting extensive testing on a functional soft magnetoelastic ball (Figures 1F and S1), several optimization experiments were performed to evaluate its two-step mechanical-to-electrical performance. A 2.6 T impulse magnetic field was applied to induce strong and stable magnetic anisotropy in the micromagnets embedded within the soft matrix. This process aligns the magnetic domains with a preferred orientation. Establishing this directional anisotropy is essential for maximizing magnetoelastic coupling, as it ensures that mechanical deformation of the sphere results in pronounced reorientation and displacement of magnetic dipoles, which, in turn, generates significant changes in the magnetic flux. The surface magnetic flux density was observed in both an undeformed state and a compressed state of 60-N load using a laboratory-built, three-axis adjustable motion platform with a Hall sensor (Figure S2). As shown in Figures 2A and 2B, compression resulted in a lower magnetic flux density, demonstrating the movement and redistribution of magnetic particles within the magnetoelastic wall, driven by the motions of the internal glass sphere (Figures 2C and 2D). The hysteresis loops further characterized the magnetization process, revealing differences before and after magnetic treatment (Note S1 in the supplemental information). Correspondingly, the scanning electron microscope (SEM) images in Figures 2E and S3 confirmed the uniform distribution of the magnetic particles within the silicone polymer as a homogeneous structure. The observed porosity of the magnetoelastic wall minimizes motion artifacts, enhancing signal stability (Note S2 in the supplemental information).[24] The performance of the MC layer was further evaluated at varying weight concentrations of

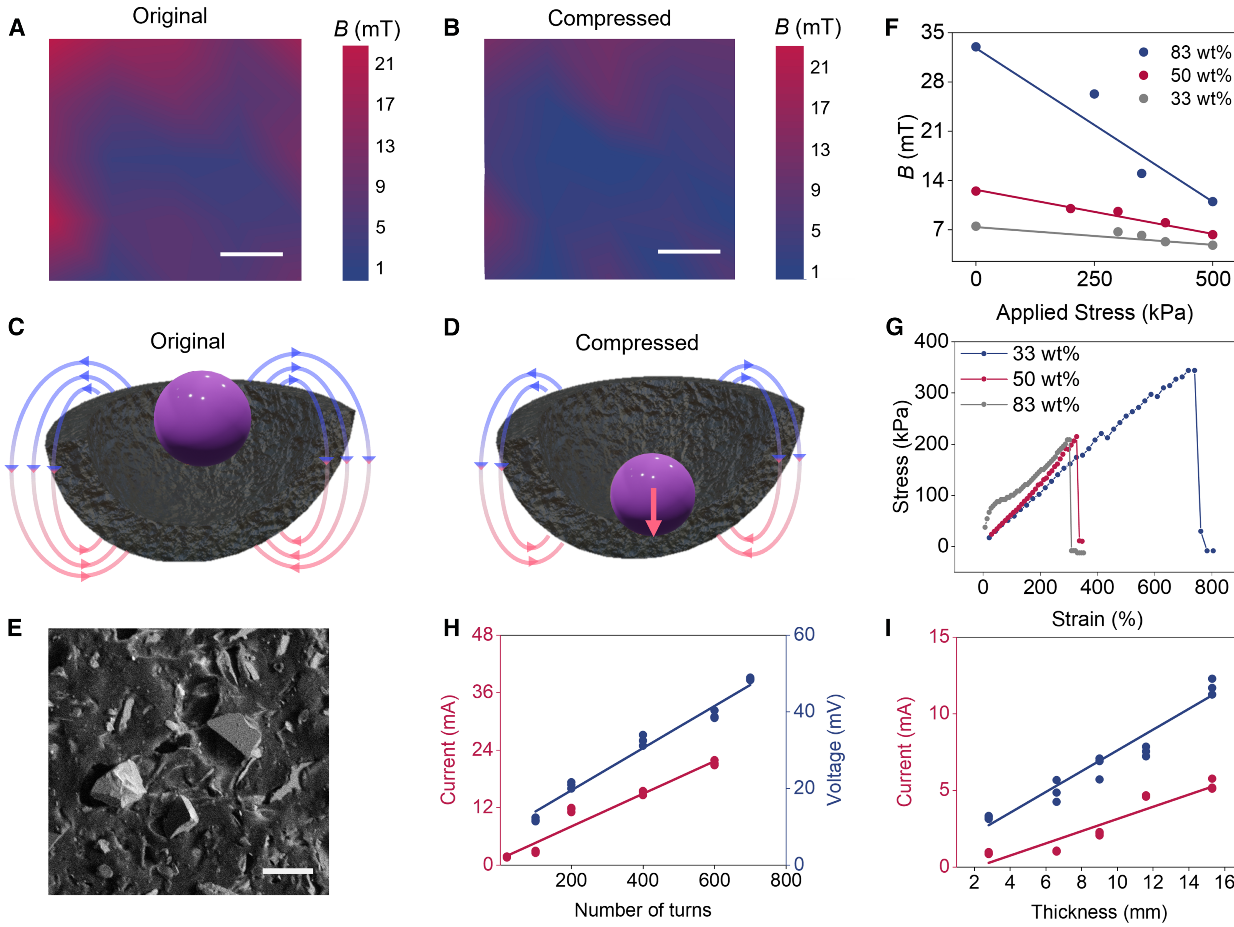

Figure 2. Evaluation of the soft magnetoelastic ball for mechanical-to-electrical conversion

(A and B) Magnetic flux density mapping of the soft magnetoelastic wall under the original condition (A) and under a uniaxial compression of 60 N (B). Scale bars: 2.5 mm.

(C and D) Illustrations of the magnetic flux density of the soft magnetoelastic ball in the original condition (C) and in the compressed condition (D).

(E) SEM image displays the distribution of magnetic particles at 83 wt% in the silicone polymer matrix. Scale bar: 100 μm.

(F) Variation in the magnetic flux under applied stress of different magnetic concentrations.

(G) Stress-strain curves of the magnetoelastic wall from different magnetic concentrations.

(H) Dependence of the current output of the soft magnetoelastic ball on the number of copper fiber turns.

(I) Electrical outputs of the magnetoelastic wall under different thicknesses.

33 wt%, 50 wt%, and 83 wt%. As seen in Figure 2F, under continuous uniaxial applied pressure from 0 to 500 kPa, the 83 wt% sample exhibited the highest magnetic field variation (21 mT), compared with 50 wt% (7 mT) and 33 wt% (2 mT) samples. The 83 wt% MC layer also achieved a strain of up to 300% (Figure 2G) and a Young's modulus of 214.53 kPa, determined using the third-order Yeoh model (Figure S4), ensuring sufficient deformation for effective signal generation. This concentration was selected for further performance testing and use in the PD application. As a trade-off, decreasing the concentration diminishes magnetic field variation, while increasing the concentration can hinder the deformability of the MC layer due to increased rigidity.

To further investigate the synergistic bonding between the magnetic particles and polymer matrix, we examined the possibility of debonding to determine the interfacial stability of the material. A cyclic actuation of 12,000 cycles was applied at parameters mimicking real-world conditions—specifically, hand-tremor-level deformations relevant to PD detection (6 Hz frequency and 12 N, which exceed the usual force exerted by a PD patient during examination). Tremor frequencies can range from 4 to 6 Hz,[46] while the grip forces in early-stage PD are between 7 and 10 N.[47] Consequently, an SEM image of the 83 wt% sample was taken after the cyclic and tensile tests. As seen by the SEM image from Figure S5A and the tensile test from Figure S5B, even at higher concentration, after multiple cycles, the debonding between the magnetic particles and polymer matrix was insignificant. After the cyclic test, the Young's modulus was measured at 181.52 kPa, the break strain at 314.8%, and the maximum stress at 136.1 kPa. These results indicate only a 15.4% reduction in Young's modulus (compared with 214.53 kPa in the original 83 wt% sample), while the strain remains minimally increased (299.7% in the original 83 wt%). Although the maximum stress decreased (208.8 kPa in the original 83 wt%), this could be due to structural relaxation. The results reveal negligible differences in tensile modulus and strain capacity before and after cycling, indicating that the mechanical integrity of the material remains stable even after prolonged use. This suggests that any potential interfacial debonding during use is minimal and does not compromise the structural or functional performance of the material.

Additionally, the size and configuration of the MI layer also influence the ability of electricity generation. The receiving coils are strategically placed in the plane perpendicular to the magnetization direction of the magnetoelastic wall, concentrated around the central axis of the sphere. This spatial arrangement is designed to maximize time-varying magnetic flux during deformation, thereby optimizing the inductive signal detected by the coils. The coupling efficiency between mechanical deformation and electrical signal output is thus enhanced by the combined effects of directional magnetization and optimal coil alignment. A coil with 600 copper fiber turns can provide adequately readable signal outputs while still maintaining sufficient internal space for the movement of the glass sphere (Figure 2H). Taking into consideration the glass sphere's movement, the thickness of the MC layer is also another important factor to consider in terms of available inner space, and hence, 6 mm provides a balance between deformation capacity and electrical output efficiency for sensing purposes (Figure 2I). Increasing the number of coil turns has a stronger effect on the signal output because more turns would capture more magnetic flux and convert the magnetic signals into electrical signals. In contrast, increasing the magnetoelastic sphere's thickness while keeping the turn count constant yielded more modest improvement in signals, as the produced magnetic flux was not converted into electrical signals to be captured by the instruments.

After optimizing the MEG layers, we proceeded to assess the overall performance of the magnetoelastic ball by analyzing the impact of the glass sphere's size and quantity on signal output. According to Figure 3A, a single glass ball provided the highest signal output. This trend can be attributed to two factors. First, in a confined space, increasing the number of glass balls reduces their available movement area. This, in turn, decreases the impact force exerted by the glass balls on the MEG. According to the wavy chain model,[25] a reduction in impact force leads to less deformation of the MEG, thereby slowing the rate of magnetic field variation and resulting in a lower output current. Second, as the number of glass balls increases, mutual collisions between them during motion dissipate part of the overall kinetic energy, further reducing the output current. While in Figure 3B, an 11-mm-diameter ball was identified as optimal, offering a balance between internal space and effective stress distribution. It is worth noting that while other configurations of ball size, number, and wall thickness could potentially achieve comparable performance, this combination demonstrated high-fidelity motion detection suitable for sensing applications.

Next, we investigated the generated electrical signals of the optimized magnetoelastic ball. The ball provided a short response time of 0.05 s at 7 Hz, an SNR of 64.6 dB at 7 Hz (Figure 3C), and a sensitivity as low as 0.95 kPa (Figure 3D). The current increased linearly with pressure in the range of 0.95 kPa to 5.67 kPa before rising more sharply, likely due to the nonlinearity of the magnetoelastic effect in the high-pressure range. The ball also demonstrated consistent output across frequencies ranging from 1 Hz to 7 Hz (Figure 3E), confirming its responsiveness to human motion. Since magnetic fields remain stable while permeating through water,[24] the magnetoelastic ball is intrinsically waterproof, allowing for stable signals in sweaty hands. Figure 3F shows the soft magnetoelastic ball submerged in water while still maintaining its original shape. A photograph in Figure S6 displays the sprayed soft magnetoelastic ball on a human hand, and Figure 3G presents the signals before and after interacting with water, further validating its waterproof capability. Additionally, the soft magnetoelastic ball exhibited adequate functionality without signal deterioration in the durability test using a calibration electrodynamic system, as evidenced by its unchanged electrical output after 12,000 cycles (Figures 3H and 3I). This experiment was designed intentionally to assess the mechanical and electrical durability of the device under a consistent, load-unload regime. In summary, the soft magnetoelastic ball holds a collection of exceptional features, including high current output, short response time, decent SNR, flexibility, durability, and waterproofness. With a prototype suitable for testing, we proceeded to examine its application in differentiating PD from healthy movements.

Application of the soft magnetoelastic ball for PD diagnostics

PD is a progressive neurological disease that adversely influences one's mobility, inducing bradykinesia, tremors, rigidity, and postural instability. Accompanying the widely accepted clinical examination of PD, focusing on the hands and fingers, the soft magnetoelastic ball provides quantitative results for PD symptoms inhering in motion-powered operation, signal-to-function conversion, and waterproof performance.

To validate the efficacy of the soft magnetoelastic ball, ten real-time examinations involving tasks performed by healthy and simulated PD[48] individuals were conducted. During these tests, the soft magnetoelastic ball was manually compressed by a human hand, therefore exhibiting signal morphologies that differed from those generated by a calibration electrodynamic system. Each task was examined repeatedly for at least 10 s using the magnetoelastic ball and recorded using a Stanford low-noise current pre-amplifier (Model SR570). The experiments were modeled in accordance with the Stanford Medicine 25, a program designed to standardize clinical examination procedures for properly diagnosing PD as well as monitoring response to treatment or progression of the disease. The Stanford Medicine 25 exam was led by Dr. Kathleen Poston, a neurologist from Stanford with expertise in movement disorders.[49] The original exam covers bradykinesia, rigidity, tremors, gait, and postural instability issues. We specifically selected tasks involving hands and fingers and incorporated the magnetoelastic ball as a quantitative measurement tool. Figure 4 describes the ten executed tasks. Furthermore, it is worth noting that the magnetoelastic ball could also be employed in future assessments as a complementary quantitative tool for a wider range of hand- and finger-related tasks.

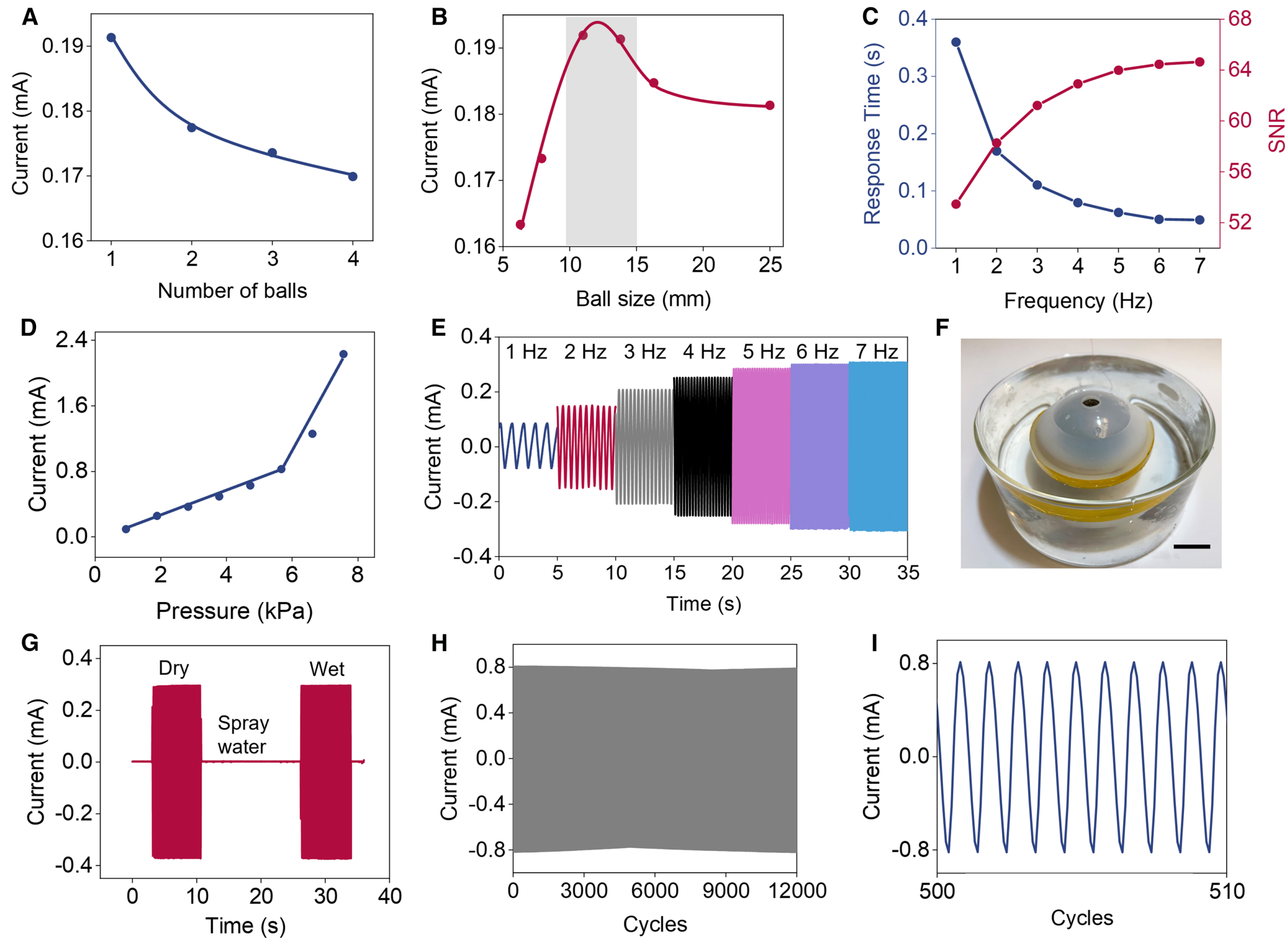

Figure 3. Examination of the soft magnetoelastic ball in a standard pressure sensing scenario

(A) Dependence of the current output of the soft magnetoelastic ball on the number of glass spheres.
(B) Dependence of the current output of the soft magnetoelastic ball on the ball's diameter.
(C) Response time and SNR under applied frequencies of 1–7 Hz.
(D) Sensitivity of the soft magnetoelastic ball.
(E) Generated current waveforms of the soft magnetoelastic ball under applied frequencies of 1–7 Hz.
(F) Photograph of the soft magnetoelastic ball submerged in water. Scale bar: 12 mm.
(G) Stability of the soft magnetoelastic ball's current output after being sprayed with water.
(H) Durability test of the soft magnetoelastic ball for 12,000 cycles.
(I) Enlarged view of the cyclic test, showing excellent output stability and repeatability.

To enhance the recognition accuracy of hand and finger gesture signals, we integrated a 1D-CNN algorithm into the system and a smartphone app (Figure 5A), utilizing the soft magnetoelastic ball as input for ten collected features of controlled condition vs. simulated PD examinations. The 1D-CNN algorithm, a member of the CNN family that has witnessed significant algorithmic advancements in recent years,[50,51] was chosen to achieve our objective due to its compact architecture and suitability for real-time, low-cost hardware implementation. A 1D-CNN typically consists of a compact three-layer configuration: a convolutional layer, a pooling layer, and a fully connected layer, each of which is capable of learning progressively abstract, higher-level representations of the input data pertinent to carrying out specific tasks and recognizing various patterns recorded by the soft magnetoelastic ball (Figure 5B). Additionally, 1D-CNN is a simple and compact configuration that only performs scalar multiplications and additions (1D convolutions), thus yielding real-time and low-cost hardware implementation.[52] During the execution of the 1D-CNN, 200 samples were multiplied and randomly divided into training and validation sets at a ratio of 8:2. The loss and accuracy function (Figure 5C) demonstrates the high efficiency of the 1D-CNN. The graph shows that, after 200 training epochs, the suggested 1D-CNN model attains high-classification accuracy and robustness. According to the confusion matrix of the true and predicted labels in the testing set (Figure 5D), which recognize healthy conditions from simulated PD conditions, our system achieves an average prediction accuracy of 98.36%.

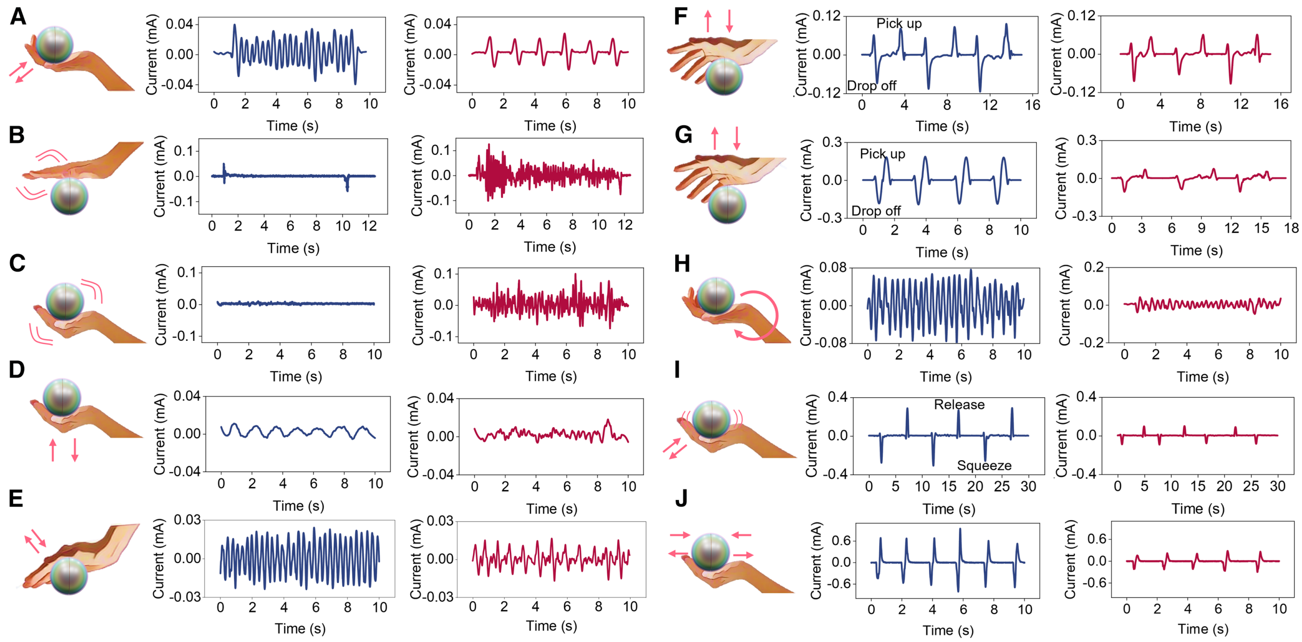

Figure 4. Electrical signals produced by the soft magnetoelastic ball using simulated PD features. Healthy (blue) vs. simulated PD (red)
(A) Opening and closing of the fingers while holding the soft magnetoelastic ball in the air at arm's length.
(B) Hand resting on the soft magnetoelastic ball, which is placed on a table surface.
(C) Holding the soft magnetoelastic ball stationary at arm's length.
(D) Moving the soft magnetoelastic ball up and down, parallel to the shoulder, while holding it at arm's length.
(E) Opening and closing of the fingers while the soft magnetoelastic ball is on a table surface.
(F) Picking and dropping the soft magnetoelastic ball for grip comparison.
(G) Picking up and dropping the soft magnetoelastic ball and repeating multiple times for speed and grip comparison.
(H) Hand pronation and supination while holding the soft magnetoelastic ball.
(I) Squeezing the soft magnetoelastic ball for 5 s and releasing.
(J) Squeezing the soft magnetoelastic ball and releasing immediately.

Following optimization of the 1D-CNN model, we integrated a user-friendly prototype smartphone application interface to enable real-time, adaptive, one-click delivery of recorded data to desired recipients. As shown in Figure 5E, after signal conditioning and data processing, the collected control and simulated PD signals were wirelessly transmitted via a Bluetooth module to a smartphone and displayed on a lab-customized application terminal in a real-time, continuous approach (Figures 5F and S7). The analyzed results were then also shown on the screen. Video S1 shows the real-time, continuous measurement of grip strength recorded by the soft magnetoelastic ball on the customized app. The smartphone application includes several unique features, such as the ability to upload collected data to a cloud database for analysis and one-click sharing of the data with clinicians via email, phone, message, or a wireless local area network. Video S2 demonstrates the options for sharing data with third-party recipients. With this wireless system, the soft magnetoelastic ball not only can assist physicians in clinical settings but also allows for personalized at-home, bedside quantification of PD progression, realizing personalized medication management in a more inclusive approach that accommodates users with diverse needs, including those who may lack visual, verbal, hearing, and writing capabilities.

DISCUSSION

Compared to other diagnostic methods, the soft magnetoelastic ball distinguishes itself in many aspects, including intrinsic humidity resistance, high current output and signal-to-noise ratio, and especially a strong commitment to inclusion and diversity. By relying solely on hand and finger movements, the soft magnetoelastic ball complements traditional clinical examination by providing quantitative analysis for physicians, without requiring users to speak, hear, see, or write. From a fundamental new perspective, the soft magnetoelastic ball operates through mechanical-to-electrical conversion by coupling the giant magnetoelastic effect and electromagnetic induction. By collecting both tremor-related pressure and timing information, the device can be used in multiple PD-related tasks, enhancing the overall assessment process.

In summary, we adopted the giant magnetoelastic effect to realize a compelling prototype for PD diagnostics using a soft magnetoelastic ball with low cost, stretchability, flexibility, and waterproofness. The ball exhibited an SNR of 64.6 dB, a response time of 0.05 s, and a sensitivity as low as 0.95 kPa. Under simulated conditions relevant to PD, we have recorded various features of hand and finger tremors collected using the soft magnetoelastic ball. With the assistance of a deep-learning

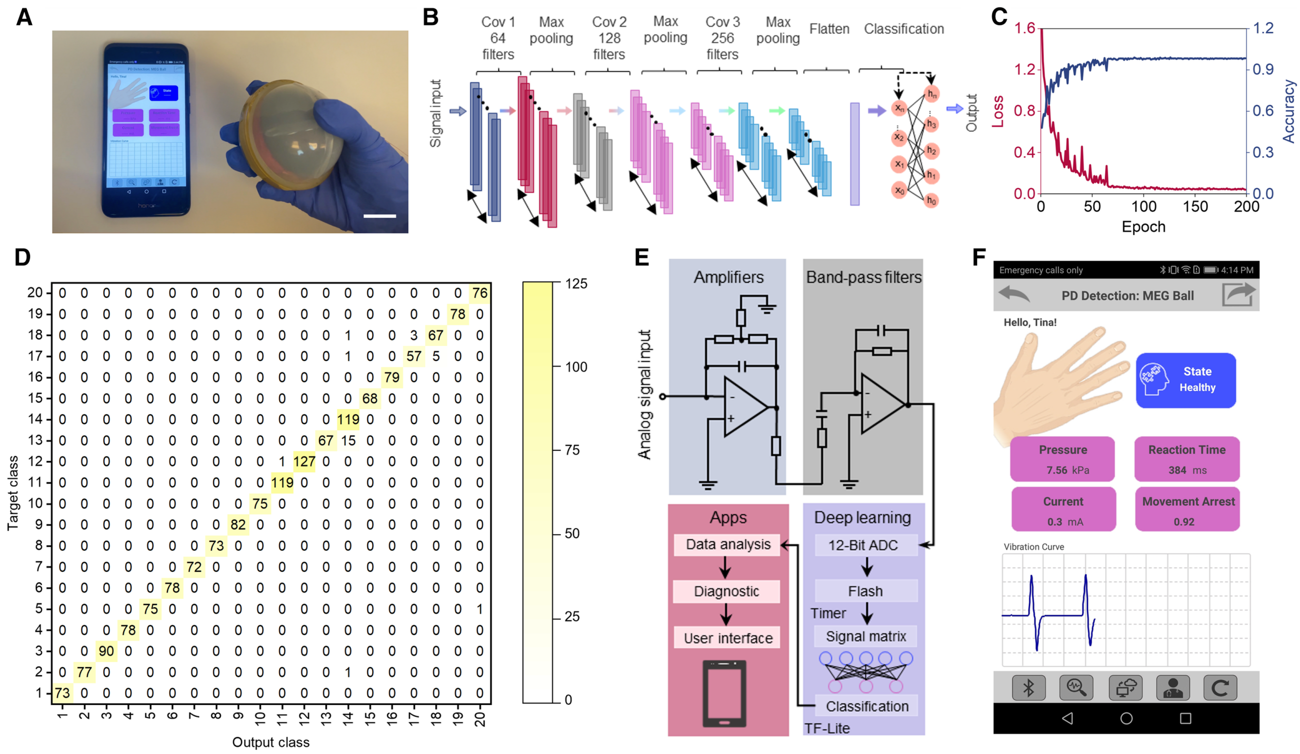

Figure 5. Demonstration of a deep-learning-assisted soft magnetoelastic ball for quantifying PD progression

(A) Photograph of a lab-customized smartphone application and the soft magnetoelastic ball. Scale bar: 12 mm.

(B) Architecture of the 1D-CNN model.

(C) Evaluation of the classification accuracy and loss function in 200 epochs.

(D) Corresponding 20 output spectra of the hand and finger movements while using the soft magnetoelastic ball as input of the 1D-CNN model. The following is the legend for classes 1–20: opening and closing the fingers while holding the soft magnetoelastic ball in the air at arm length, from control (1) and simulated PD (2); hand resting on the soft magnetoelastic ball placed on a table surface, from control (3) and simulated PD (4); holding the soft magnetoelastic ball stationary at arm's length, from control (5) and simulated PD (6); moving the soft magnetoelastic ball up and down, parallel to the shoulder, while holding it at arm's length, from control (7) and simulated PD (8); opening and closing of the fingers while the soft magnetoelastic ball is on a table surface, from control (9) and simulated PD (10); picking up and dropping the soft magnetoelastic ball for grip comparison, from control (11) and simulated PD (12); picking up and dropping the soft magnetoelastic ball and repeating multiple times to test speed comparison, from control (13) and simulated PD (14); hand pronation and supination while holding the soft magnetoelastic ball, from control (15) and simulated PD (16); squeezing the soft magnetoelastic ball for 5 s and releasing, from control (17) and simulated PD (18); and squeezing the soft magnetoelastic ball and releasing immediately, from control (19) and simulated PD (20).

(E) Schematic of the in-lab circuit system and the deep-learning-assisted PD progression monitoring system, including the soft magnetoelastic ball, an amplifying circuit, a band-pass filter circuit, a machine learning algorithm, a Bluetooth module, and a lab-developed smartphone application or computer interface that sends data to physicians with one touch.

(F) Screenshot of the lab-customized smartphone app.

algorithm, 1D-CNN, an average prediction accuracy of 98.36% was achieved in pattern recognition between the healthy and the simulated PD states.

Despite the promising outcomes, this study remains a preliminary investigation with inherent limitations. While the current system has been validated under controlled laboratory experiments, further investigations, including human studies with PD patients, are necessary to advance the soft magnetoelastic ball toward practical deployment in real-life scenarios. Looking ahead, integrating big-data analytics from clinical datasets could enhance diagnostic accuracy and enable personalized approaches, ultimately aiming to translate this technology into a commercially viable tool to assist physicians and patients in diagnosing PD. By addressing both the potential and current limitations of this approach, this study lays the groundwork for future translational research, demonstrating how the soft magnetoelastic ball can support clinical testing of PD and contribute to more objective diagnostic assessments. In addition, this strategy emphasizes diversity and inclusion in healthcare, promoting a more comprehensive and personalized model of patient care.

METHODS

Fabrication of the soft magnetoelastic ball

The soft magnetoelastic ball was fabricated using commercially available materials. The outermost stretchable silicone structural support and the innermost glass sphere were purchased commercially on Amazon.com. The magnetoelastic wall was fabricated using Ecoflex 00-30 part A and Ecoflex 00-30 part B with a weight ratio of 1:1. NdFeB micromagnets were purchased from Neo Magnequench (MQP-B+ (−150 m), 20441-086). The odorless, powder product came in the form of crushed ribbon powder and comprised 19.1 wt% of neodymium, 6.3 wt% of praseodymium, 0.9 wt% of boron, and 73.7 wt% of iron. These bonded neo powders were produced using rapid solidification techniques—first by jet casting to produce ribbons, followed by jet milling into a platelet or flake-shaped morphology. The uncoated particles have a density of 7.59 ± 0.20 g/cm^3 and an apparent density of ~2.7 g/cm^3. The powdered particles have a melting point of 1648.89°C. The particle size distribution is > 140 mesh (107 μm) < 2 wt%, > 170 mesh (89 μm) < 20 wt%, and < 325 mesh (45 μm) < 35 wt%. Magnetic properties of the powder included a residual induction of 889 mT, a maximum energy product of 119 kJ/m^3, and an intrinsic coercivity of 725 kA/m. No surface coatings were applied to the particles prior to forming the magnetoelastic shell.

Weight concentrations of 33%, 50%, and 83% were mixed vigorously with Ecoflex using a stirring rod for about 10 min to introduce air bubbles and produce a porous structure. The mixture was then poured into a 3D spherical mold, and a smaller spherical mold was introduced on top to create a hollow shell. This step was repeated twice to produce two hemispheres, which were subsequently joined to form a complete sphere. This platform was set in an ultrasonic device for sonication to induce uniform particle dispersion, followed by rapid thermal curing at 70°C and 30% in the oven (Thermo Fisher Scientific) for 4 h to preserve the uniformity of the dispersion. The fast curing and high viscosity prevent the particles from settling at the bottom of the mold. The magnetoelastic hemispheres were magnetized using a commercial impulse magnetizer (IM-10-30, ASC Scientific). A single high-intensity magnetic pulse with a peak field strength of approximately 2.6 T was applied, with a pulse duration of ~3 ms. Conductive copper wires were wound with different numbers of turns for testing. The coil is BNTECHGO 38 AWG magnet wire (enameled copper wire; enameled magnet winding wire: 0.0039 in diameter; 1-spool coil; red; temperature rating: 155°C). For the performance characterization and PD application, the 600-round coil had a resistance of 105.4 Ω. Then, the completed coil layer was inserted between the sphere, and glass spheres were positioned in between to complete the soft magnetoelastic ball.

Structural characterization of the soft magnetoelastic ball

Structural characterization of the MC layer was conducted by scanning electron microscopy (Zeiss supra 40VP). The micro-CT image of the MC layer was scanned at 80 kVp/140 μA with 500 ms exposure using a μCT scanner (HiCT) developed by the Crump Institute for Molecular Imaging at UCLA. The 2D magnetic flux density mapping was achieved using a digital Gauss meter (TD8620, Tunkia), incorporated into a custom-built, three-axis adjustable motion platform. This platform consists of a mechanically stable, motorized X-Y-Z positioning stage that allows precise spatial control of the measurement probe. The TD8620 digital Gauss meter is mounted at the tip of a customized probe holder and is capable of capturing axial components of the magnetic flux density. The stage is controlled by a microcontroller-based driver system that enables programmable scan patterns with tunable step size and scanning speed. The scanning resolution of the platform can be adjusted down to a minimum of ~0.01 mm/s, allowing high-density spatial sampling across the surface of the magnetoelastic structures. Uniaxial stress was applied to a magnetoelastic film of size 1 cm × 1 cm with an axial scanning resolution of 2.5 mm.

The stress-strain curves were characterized by a universal mechanical analyzer (Instron 5564) moving at a rate of 5 mm/min under a uniaxial tensile test configuration. The samples were stretched until failure (breakage). The Young's modulus was calculated using the third-order Yeoh model due to the curve's nonlinearity. The Yeoh model was chosen due to the material's high strain, soft composite, and nonlinear stress-strain data under uniaxial tensile test.

Electrical performance measurement of the soft magnetoelastic ball

The voltage and current signals of the soft magnetoelastic ball were measured using a Stanford low-noise voltage pre-amplifier (Model SR560), a Stanford low-noise current pre-amplifier (Model SR570), and a Keithley Model 6514 programmable electrometer. For any current outside of the equipment's input limit, a series resistance box was employed to reduce the current. The actual output current was calculated and derived from the measured reduced current. The effect of coil turns was examined by varying the number of turns while maintaining a constant magnetoelastic wall thickness of 6 mm. Conversely, the effect of magnetoelastic wall thickness was investigated by varying the thickness while keeping the coil fixed at 10 turns. The loading parameters for the standard output characterization regarding the number of turns and thickness experiments consist of a displacement of 10 mm and a frequency of ~3 Hz. The loading parameter for the performance optimization experiments, varying the internal ball number and size, was 1 Hz. The standard output characterization, performance optimization, and cyclic performance for the durability of the soft magnetoelastic ball were validated by a calibration electrodynamic system, consisting of a function generator (AFG1062, Newark), a linear power amplifier (PA-151, Labworks Inc.), and an electrodynamic transducer (ET-126HF, Labworks Inc.). The output performance properties, such as SNR, response time, varying frequencies, and sensitivity, were also characterized by the calibration electrodynamic system. The response time was calculated using the time between when the current was 0 mA and when the current was optimal.

Deep learning tool for differentiating PD tremors and normal movements

A three-layer 1D-CNN was built to extract data features and automatic recognition of 20 control and simulated PD inputs combined. The samples were divided randomly into training and testing sets of 4:1. 80% of the data was used to train the proposed model, while 20% was used to validate the model parameters. To examine the deep-learning effect, the classification accuracy and loss function were evaluated throughout each of the 200 training epochs, achieving highly simulated PD recognition patterns with the assistance of the

1D-CNN algorithm. Thereafter, classification analysis was conducted to determine the classification accuracy and loss function.

Design the circuitry and user interface for wireless diagnosis of PD symptoms

The PD system includes a soft magnetoelastic ball, whose signals were first amplified using an amplifier and filtered using a band-pass filter to remove noise. Then, the analog signals were converted to digital signals for subsequent processing using a microcontroller (Arduino UNO), with a trained 1D-CNN model developed in Python with TensorFlow Lite (TF-Lite) backend. The classification results were sent to a smartphone. The lab-customized Android application for PD progression monitoring was developed using the MIT AI2 Companion. It could continuously process recorded hand and finger gesture signals and provide additional assessment and analysis in real time. The data were transmitted via Bluetooth (HC-05). The data could also be stored in a cloud database and shared with desired recipients via email or message.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Jun Chen (jun.chen@ucla.edu).

Materials availability

This study did not generate any new unique reagents.

Data and code availability

All data reported in this paper will be shared by the lead contact upon request. This paper does not report original code.

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AUTHOR CONTRIBUTIONS

Conceptualization, T.T., J.X., and J.C.; methodology, T.T. and J.X.; investigation, T.T., J.X., X.Z., and S.C.; formal analysis, T.T., J.X., S.C., and Z.D.; data curation, X.X.; visualization, T.T., J.X., and J.Y.; writing—original draft, T.T. and J.X.; writing—review and editing, T.T., J.X., Z.D., and G.C.; funding acquisition, J.C.; resources, J.C.; and supervision, J.C.

DECLARATION OF INTERESTS

The authors declare no conflict of interest.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this manuscript, the authors used ChatGPT only to polish the language. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

SUPPLEMENTAL INFORMATION

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CELBIO, Volume *2*

Supplemental information

A soft magnetoelastic ball for self-powered Parkinson's disease diagnosis

Trinny Tat, Jing Xu, Xiao Xiao, Songyue Chen, Zhaoqi Duan, Xun Zhao, Guorui Chen, Junyi Yin, and Jun Chen

Supplemental Figures

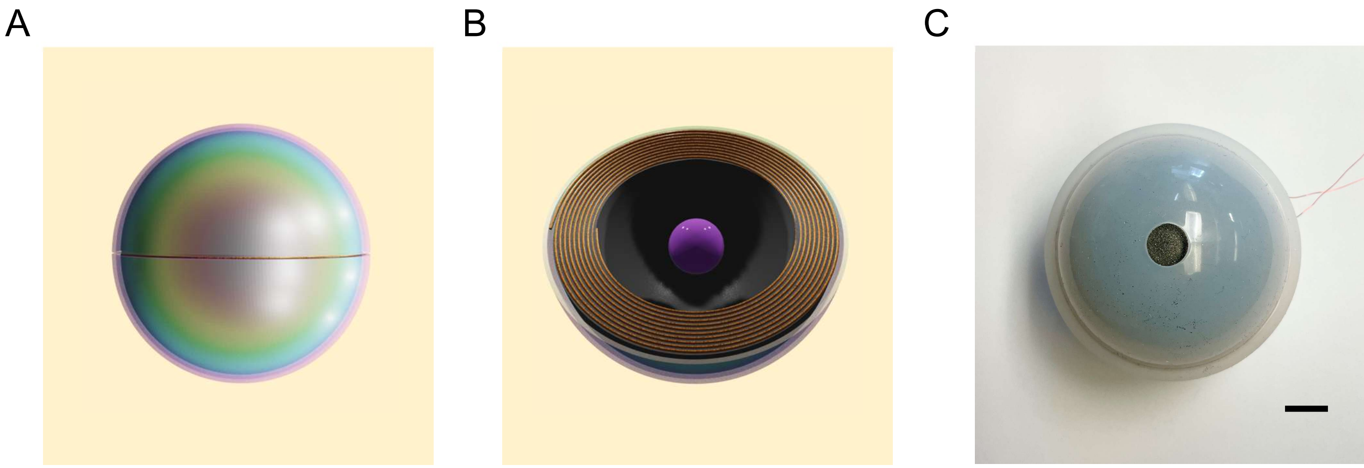

Figure S1. Soft magnetoelastic ball.

(A) Full view.
(B) Internal view.
(C) Photograph of the overall device. Scale bar: 12 mm.

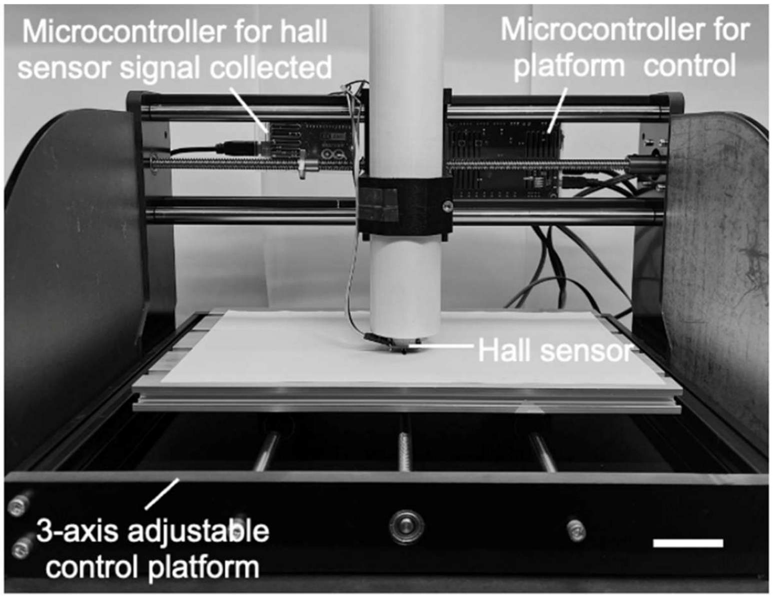

Figure S2. Experimental setup of the magnetic flux density mapping measurements. Scale bar: 3 cm.

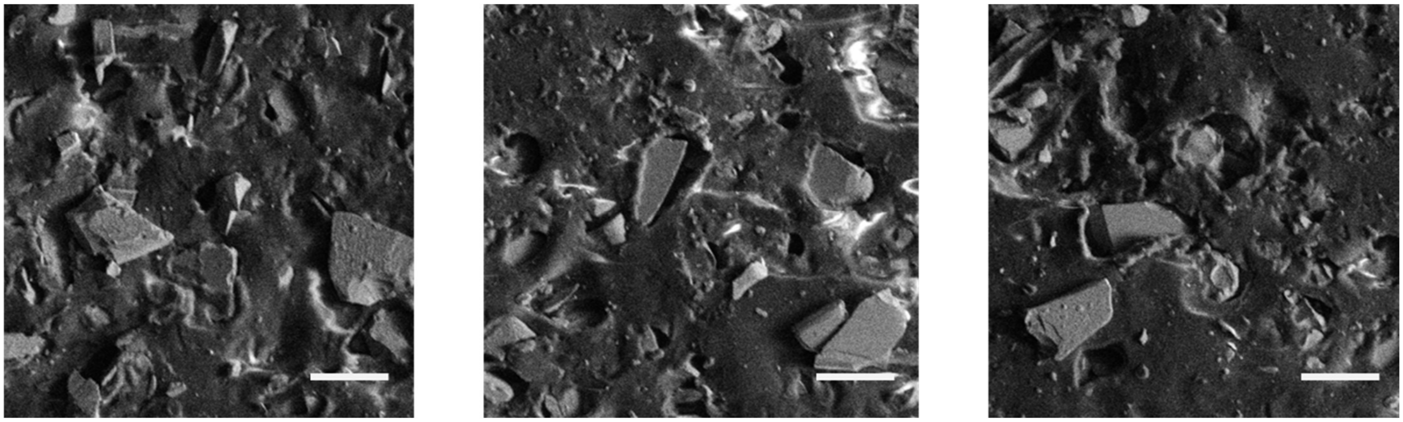

Figure S3. SEM images from different regions of an 83 wt% magnetoelastic polymer. Scale bars: 100 μm.

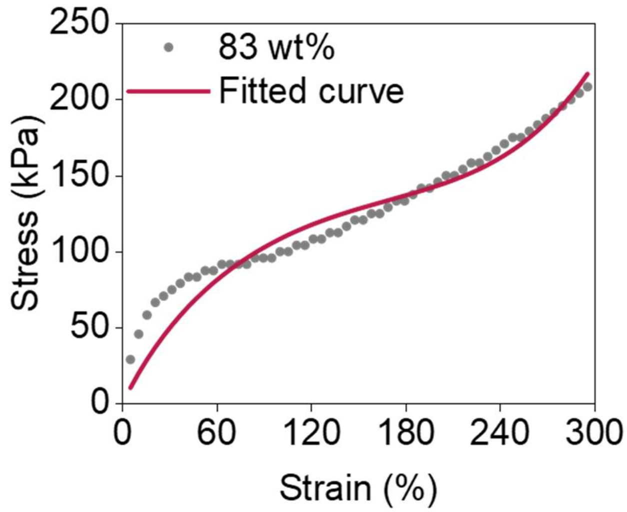

Figure S4. Fitting of Yeoh model for 83 wt%, Young's modulus of 214.53 kPa.

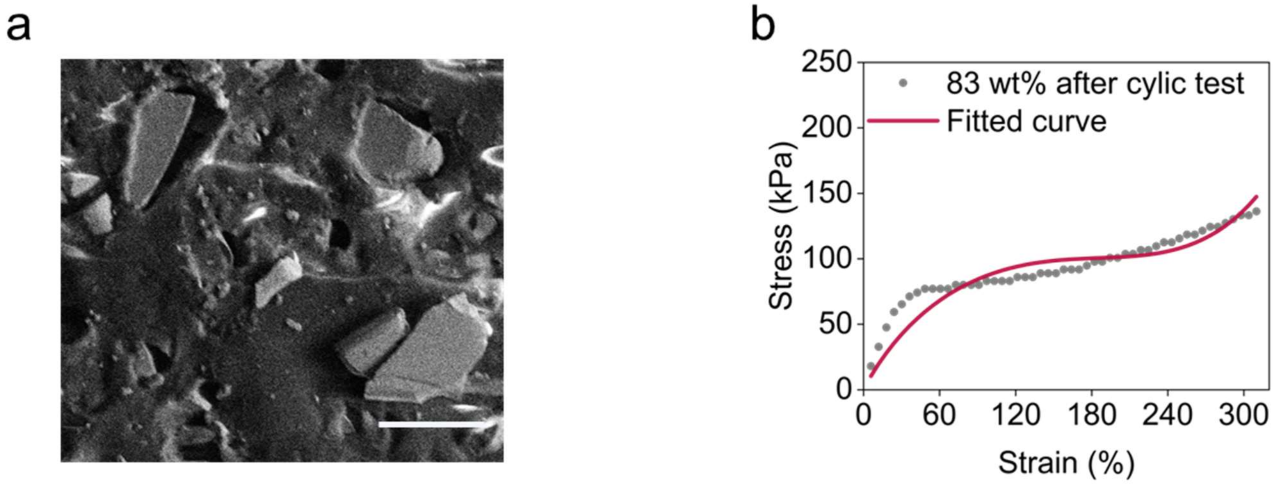

Figure S5. Observation of 83 wt% after a cyclic test of 12,000 cycles at 12 N and 6 Hz. (A) SEM image. Scale bar: 100 µm. (B) Fitting of Yeoh model after 12,000-cycle durability test; Young's modulus of 181.52 kPa**.**

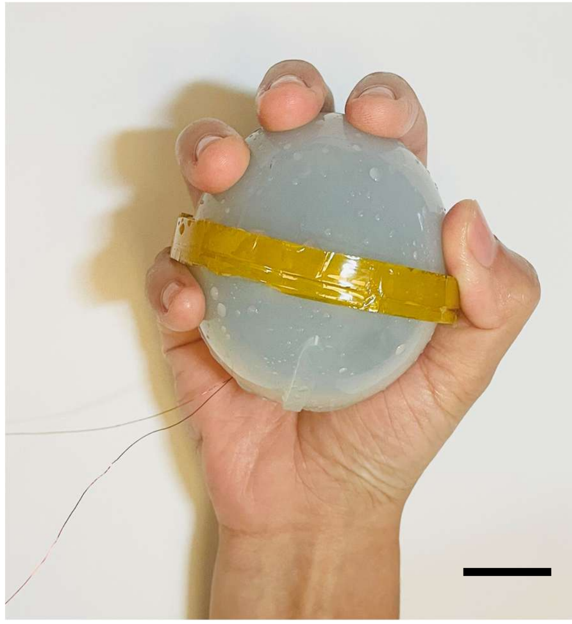

Figure S6. Photograph of a wet intelligent magnetoelastic ball on the human hand. Scale bar: 12 mm.

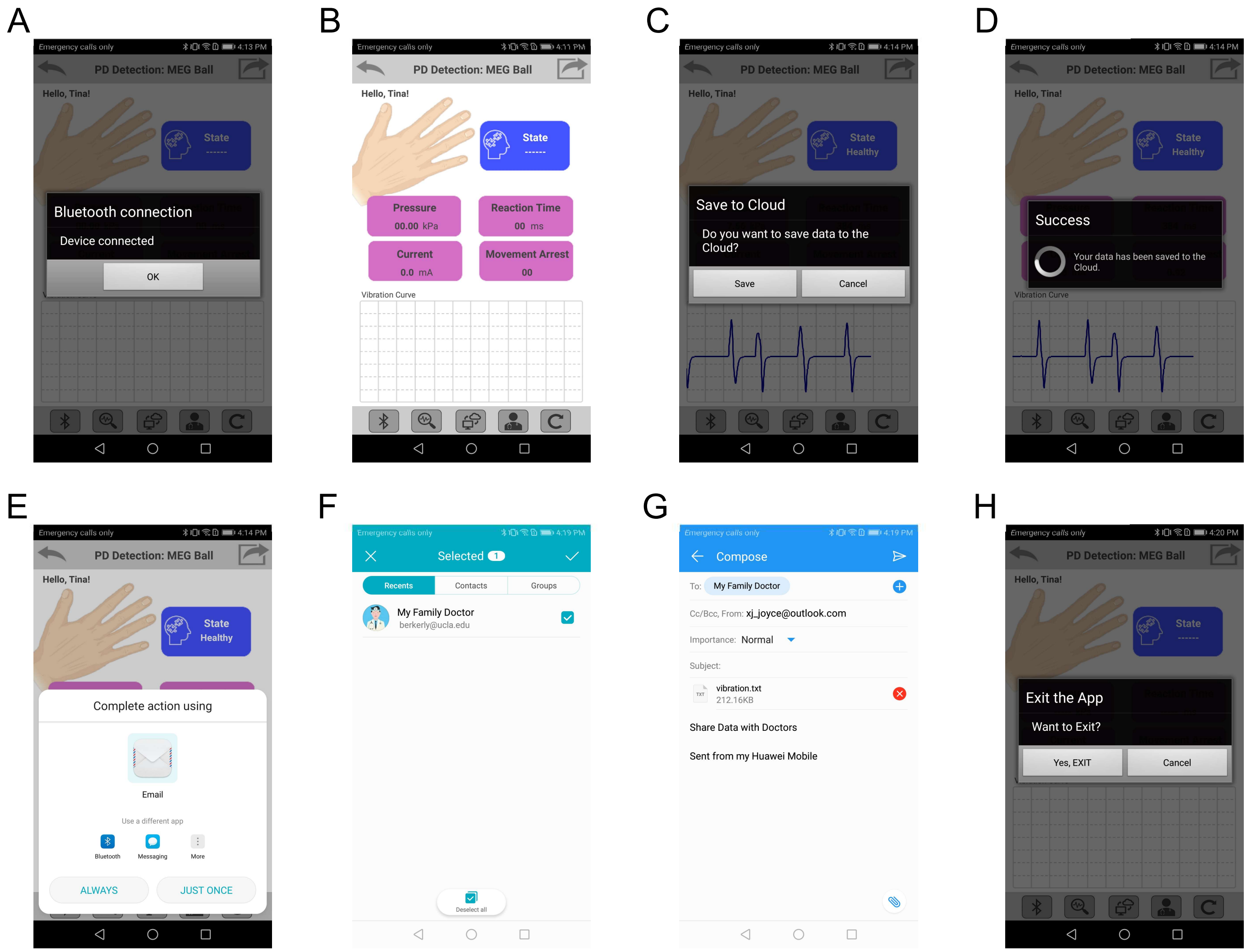

Figure S7. Screenshots of the cellphone APP.
(A) Connecting to Bluetooth.
(B) Initial interface.
(C) Saving to the cloud after collecting data.
(D) Successfully saving to the cloud.
(E) Sharing options.
(F) Selecting a desired recipient.
(G) Emailing the desired recipient.
(H) Exiting the cellphone APP.

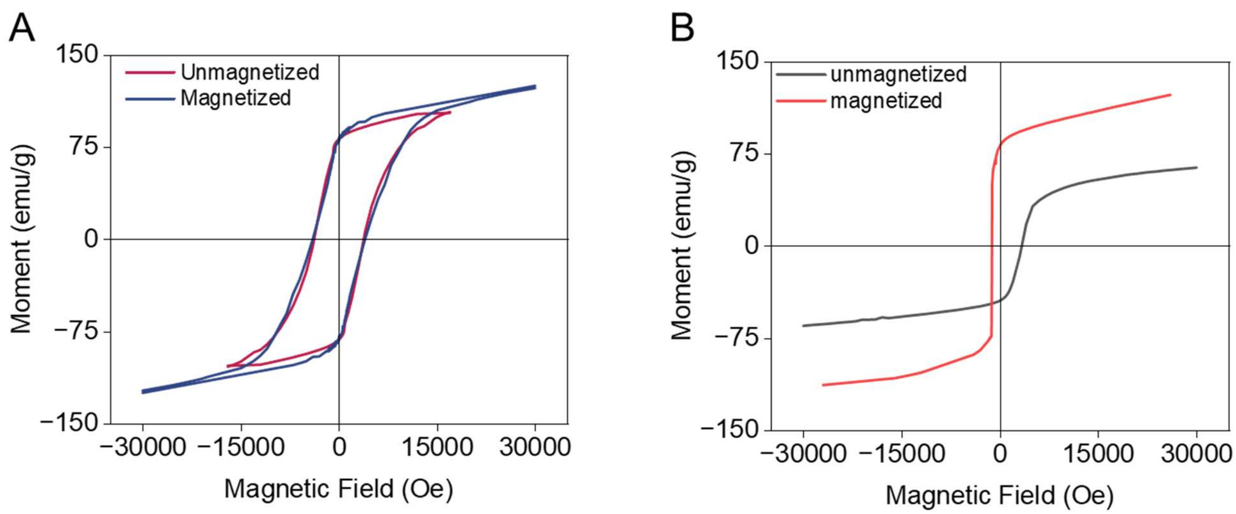

Figure S8. Hysteresis loop characterizing the magnetization behavior of the micromagnets in the soft magnetoelastic film.
(A) 83 wt% with a porosity of 0.191.
(B) 83 wt% with a porosity of 0.412.

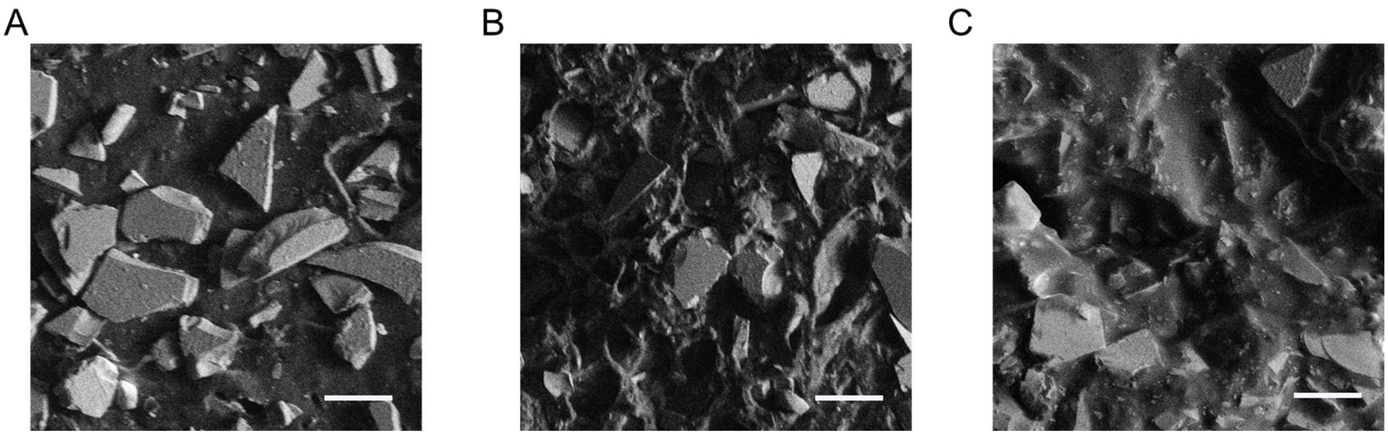

Figure S9. Morphology of the 83 wt% magnetoelastic walls with different porosities using SEM images.
(A) 0 porosity. Scale bar: 100 μm.
(B) 0.191 porosity. Scale bar: 100 μm.
(C) 0.412 porosity. Scale bar: 100 μm.

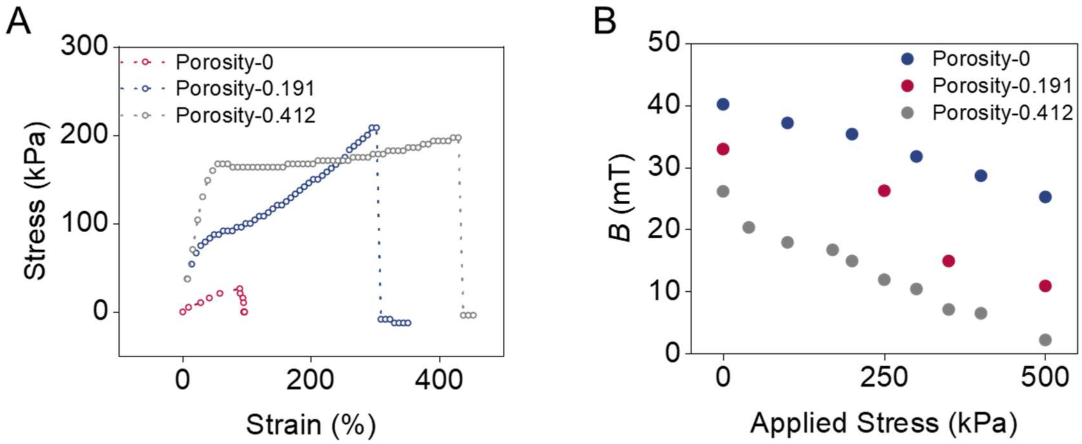

Figure S10. Mechanical and magnetic properties of the 83 wt% magnetoelastic walls with different porosities
(A) Stress-strain curves of the 83 wt% magnetoelastic walls with different porosities.
(B) Magnetic flux variation of the 83 wt% magnetoelastic walls with different porosities.

Supplemental Notes

Note S1. Characterization of the magnetization before and after magnetic treatment.

It is important to note the significant differences in the magnetization processes between the impulse magnetizer and the superconducting quantum interference device (SQUID) used to generate magnetic hysteresis loops.

The impulse magnetizer applies a high magnetic field pulse lasting only a few milliseconds, as mentioned in the Experimental Procedure. Due to this extremely short duration, the magnetic materials experience minimal magnetostrictive deformation (i.e., elongation or contraction along the magnetization axis), as the mechanical response time of the material—especially in soft matrices like Ecoflex—likely exceeds the duration of the magnetic pulse.

In contrast, SQUID magnetometry involves a gradual and quasi-static sweep of the external magnetic field, often taking several minutes per full cycle, and reaching up to 30,000 Oe—comparable in magnitude to that of the impulse magnetizer. This slower, controlled variation in magnetic field produces a continuous and well-defined magnetic hysteresis loop that characterizes key properties such as coercivity (x-intercept) and remanent magnetization (y-intercept). According to the Jiles–Atherton model of hysteresis, an ideal magnetic system should exhibit a symmetric loop, with equal and opposite coercivity and remanent magnetization in the forward and reverse directions.

As shown in Figure S8A, although impulse magnetization slightly reduces the saturation magnetization of dense magnetic composites, the coercivity and remanent magnetization remain nearly unchanged. This suggests that impulse magnetization does not induce significant structural or magnetic alterations in the dense, less porous system.

However, the behavior differs notably from the more porous MC layer. As illustrated in Figure S8B, impulse magnetization increases the remanent magnetization, reduces coercivity, and even enhances the saturation magnetization. This behavior indicates that magnetic particles become more aligned after impulse treatment in porous systems. According to our prior research,[1,2] porosity softens the mechanical matrix but also enhances the magnetomechanical coupling coefficient, making the system more responsive to magnetic field-induced deformation.

During impulse magnetization, the magnetostriction effect becomes more apparent. The steady-state strain induced by magnetostriction in each magnetic particle is governed by:

$$e = \frac{3}{2}\lambda_s(m^2 - \frac{1}{3}) \tag{1}$$

Where e is the magnetostrictive strain, $\boldsymbol{\lambda_s}$ is the saturation magnetostriction coefficient, and m is the directional cosine of magnetization relative to the strain axis (which depends on the external magnetic field). While $\boldsymbol{\lambda_s}$ remains constant for a given material, porous systems allow greater overall strain expression due to their reduced mechanical constraint. In other words, the same internal magnetostrictive forces result in more measurable elongation in porous composites than in dense ones. This also reflects the group effect on the chain alignment of magnetic materials [S2].

This enhanced mechanical responsiveness can lead to magnetic artifacts during SQUID measurements. SQUID detects total magnetic flux through pick-up coils and is sensitive to small vertical movements or mechanical vibrations of the sample. In porous systems, magnetostrictive elongation during field sweeps may cause vertical displacement, resulting in flux instability and non-closure of the magnetic loop during cycling. Consequently, only a partial magnetization–demagnetization–magnetization cycle is shown to accurately represent the SQUID results for porous samples.

Note S2. Characterization of porosity of the magnetoelastic wall.

To characterize the various porosity levels of the magnetoelastic wall, two additional magnetoelastic walls with different porosity were prepared. Equation (2) was used to calculate porosity.

$$\phi = 1 - \frac{\rho_{bulk}}{\rho_{sample}} \tag{2}$$

Where ϕ is the porosity, ρ_{bulk} is the bulk density, and ρ_{sample} is the density of the non-porous, solid sample. The porosity-0 sample was prepared by heating the uncured mixture in a vacuum oven. Vacuum degassing was not performed on the porosity-0.191 sample in order to introduce air bubbles for a porous structure. The porosity-0.412 sample was prepared by adding an extra 5 wt% of NH_4HCO_3 (Sigma Aldrich) to the mixture with 83 wt% of micromagnet concentration. The SEM images for these magnetoelastic walls are shown in Figure S9.

We also graphed the stress-strain curves of the different porosities (Figure S10A) and calculated the Young's modulus of each graph by fitting to the third-order Yeoh model. Porosity of 0 has a Young's modulus of 311.89 kPa, porosity of 0.191 with 214.53 kPa, and porosity of 0.412 with 48.16 kPa. Furthermore, we observed that the magnetic flux variation increases with porosity (Figure S10B) although the sample with 0.412 porosity exhibited the lowest magnetic density before compression, likely due to large pores yielding less magnetic particle interactions.

These trends are consistent with previous literature report [S1,S3]. Higher porosity reduces the stiffness of the magnetoelastic wall and still produces a readable magnetic signal sufficient for a sensing device. However, a large amount of porosity disrupts the polymer matrix network and the particle alignment, showing less stretchability and possible cracks inside the matrix. Consequently, we selected the normal 83 wt% magnetoelastic wall of 0.191 porosity as it strikes a balance between mechanical compliance and magnetic signal intensity.

Supplemental References

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